

Review retrospective:
*the scalar Taylor bound — a clean first order, and a
second-order constant the file under-delivers*

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Abstract

A soundness review of `Laplace/ScalarBound.lean`: the elementary scalar Taylor-remainder bounds on e^{-z} that feed the integral-remainder machinery. The first-order bound is clean. The second-order bound is the **first genuine statement-versus-claim mismatch of the laplace review marathon**: the exported headline proves a constant $1/2$ where its own section intro — and the true Taylor remainder — give the sharp $1/6$. Per the review-tide rule this is reported, not rewritten, and escalated to the user. One dead `open` was landed alongside.

1 Setting

The scalar substrate beneath `IntegralRemainder`: it proves $|e^{-z} - (1 - z)| \leq (z^2/2) \max(1, e^{-z})$ (first order) and the second-order analogue, the former consumed directly by the perturbative-remainder integrand estimate. A pure file of elementary calculus inequalities — no integrals.

2 What was reviewed

The first-order chain ($1 - z \leq e^{-z}$; nonnegativity; the $z \geq 0$ and $z \leq 0$ case bounds; the absolute headline `abs_exp_neg_sub_one_add_le`) and the second-order chain (the $z \geq 0$ and $z \leq 0$ branches and the absolute headline `abs_exp_neg_sub_one_add_sub_half_sq_le`), checked constant-by-constant against the Taylor remainders.

3 Fidelity / soundness findings

First order: clean. Every constant and case bound matches: $z \geq 0$ gives $z^2/2$, $z \leq 0$ gives $(z^2/2)e^{-z}$, and the headline assembles them with the $\max(1, e^{-z})$ factor and the correct $1/2$.

Second order: the exported constant is loose by a factor of three. The $z \leq 0$ branch `exp_neg_sub_one_add_sub_half_sq_le_half_cube_mul_exp_neg_of_nonpos` proves $e^{-z} - (1 - z + z^2/2) \leq (-z)^3/2 \cdot e^{-z}$, and the exported headline `abs_exp_neg_sub_one_add_sub_half_sq_le` proves $\leq |z|^3/2 \cdot \max(1, e^{-z})$ — but the section intro advertises, and the $z \geq 0$ branch correctly proves, the sharp $1/6$. And $1/6$ is genuinely achievable on $z \leq 0$: with $w = -z \geq 0$, the Lagrange remainder gives $e^w - (1 + w + w^2/2) = e^\xi w^3/6 \leq e^w w^3/6 = (|z|^3/6)e^{-z}$. So the Lean settles for a bound three times looser than both the mathematics and its own documentation. (A companion docstring also cross-references a non-existent declaration name in the same section.)

This is a *fidelity* finding, not a math error: the $1/2$ bound is true. But it does not match the claimed/intended $1/6$. The review-tide rule is unambiguous here — **do not silently rewrite the mathematics, and do not quietly correct the docstring down to $1/2$** (which would erase the very signal that the Lean under-delivers). It is recorded, surfaced to the user, and left for them to resolve: either strengthen the $z \leq 0$ branch and the headline to the sharp $1/6$ (a re-proof — a forward tide), or consciously accept $1/2$ and correct the section intro/docstrings accordingly. After ~ 50 clean laplace reviews this is the first such escalation.

4 Simplifications applied

One dead open — and a triage the build corrected both ways. `open Real` is dead (every `Real` name is qualified, including the lemma names `Real.add_one_le_exp`, `Real.hasDerivAt_exp`, `Real.exp_pos`); removed, build green. Instructively, my pre-build triage had *both* opens wrong: I guessed `open Set` was dead (it is live — bare `Ici` in the half-line case splits, which my interval grep had simply not listed) and hedged that `open Real` might be live (it is dead). Removing `open Set` broke the build with `Unknown identifier Ici`; removing `open Real` did not. The build is the arbiter, for the reviewer’s grep and the author’s intuition alike.

5 Disagreements and open questions

The second-order constant (above) is the open question for the user. GPT also listed several `*_nonneg` locals as “syntactically unreferenced” but hedged that automation may consume them implicitly; on this `positivity/nlinarith`-heavy file that risk is real (the same implicit-consumption trap as the `QuarticBoundedPrior` review’s `ha/htail_nn`), so they were left report-only for a build-arbitrated hygiene pass rather than removed speculatively.

6 Lessons

- **A true-but-loose constant is a fidelity finding, not a free docstring fix.** When a theorem proves $1/2$ and its docstring claims $1/6$, the green build proves only that $1/2$ holds — it says nothing about which number was *intended*. Correcting the docstring down to $1/2$ would silently ratify the weaker theorem; strengthening the theorem to $1/6$ is new mathematics. Either way it is the user’s call, and the reviewer’s job is to surface the gap, not close it.
- **An interval grep is only as good as its list of interval names.** `Ici` was live and unlisted; the “`open Set` is dead” guess was wrong because the grep never looked for it. The build caught it instantly — which is exactly why the open-removal verdict is always build-arbitrated, never grep-final.

7 Follow-ups

- **User decision required:** the second-order $1/6$ -vs- $1/2$ constant. If $1/6$ is wanted, a forward tide re-proves the $z \leq 0$ branch (Lagrange remainder $e^\xi w^3/6$) and propagates to the headline; the cross-reference and section intro then become consistent.
- **Batched-hygiene tide:** build-arbitrate the `*_nonneg` dead-local candidates; narrow `import Mathlib`; extract the duplicated $z \leq 0$ nonnegativity proof.